

When a Control Does Not Control: Mechanism-Derived Nulls for a False Positive in p -adic ECDLP Cryptanalysis

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Abstract

Automated pipelines that fan out many structural probes against a hard problem and surface the most anomalous are false-positive engines, in two distinct ways: moderate separations ($\sim 5\sigma$) are cheap under enough multiple probing, and—more dangerously, as here—a *real* public structure tested against the *wrong null* produces apparent separations that grow without bound. This paper is a methodological case study of one such false positive, fully dissected. In a systematic negative exploration of p -adic lifting attacks on the elliptic curve discrete logarithm problem (ECDLP), a Gowers U^2 / Fourier analysis of the Teichmüller lift error $\delta(k)$ produced a persistent elevation, growing from ~ 3 to ≈ 44 standardized units with the curve order, that a standard value-permutation control certified as “real, index-ordered structure.” We show this was a *control that cannot separate the hypotheses it was meant to test*. Mechanism-derived nulls—one per candidate explanation—localize the entire signal to a single exact, public identity, the antisymmetry $\delta(k) + \delta(n - k) = [n]\tau(G)$: a positive control carrying *only* that reflection reproduces the elevation and its growth, a converse control (a genuine but non-odd section, built by kernel offsets) collapses it at full sequence length, and a sham control (an odd kernel offset) leaves it intact. We make the mechanism analytic: the reflection turns each Fourier mode into an *improper* complex Gaussian, and a pseudo-variance identity classifies the U^2 mass inflation of any linear index constraint, with the reflection attaining the maximal factor $3/2$ and a closed-form $\approx 0.427\sqrt{L}$ growth law that an auditor can apply directly. An inertness theorem—an explicit instance, in lift-error language, of a principle of Silverman and of Gadiyar–Padma, stated for an arbitrary abelian-group extension—explains why every such anomaly on these curves must reduce to artifact-or-identity for $\gcd(n, p) = 1$, and the boundary is sharp: at $n = p$ the section ambiguity is annihilated and the surviving invariant is Smart’s attack. The experiments are diagnostic, not an empirical security evaluation of deployed curves; the reach to large-prime ECDSA-type curves is through the theorem’s generality, not through cryptographic-scale experiments. The transferable contribution is the *Mechanism-Resolved Null Protocol*: a control is valid only against a named alternative hypothesis, nulls must be derived from mechanisms, a positive control must reproduce any surviving signal, a converse control should break it, and a structural reason for inertness beats an accumulation of passed tests.

*Correspondence: william.todd.trevena@gmail.com. Code, data, and a per-phase index: <https://github.com/wtrevena/ecdsa-lift> (release tag `paper-v4`; an archival DOI will accompany the camera-ready).

1 Introduction

A recurring temptation in cryptanalysis is to lift a hard problem into a richer setting—a p -adic ring, a formal group, a higher-genus Jacobian—hoping that structure invisible over $\mathbb{F}_p$ becomes visible and exploitable. For the ECDLP this hope is well mapped: Silverman’s “four faces of lifting” [2, 1] catalogues the local/global $\times$ torsion/non-torsion strategies and the obstruction each meets, and the only p -adic local lift that ever broke an ECDLP is Smart’s attack on anomalous curves [4, 5, 6], where the group order equals the characteristic.

Automating such explorations sharpens an old hazard. A pipeline that computes many encodings of a lift invariant and reports the most unusual will, after enough nulls, return an “anomaly,” and the analyst must decide whether it is signal. We conducted a long, deliberately exhaustive negative exploration of the canonical-lift attack surface for ordinary prime-order curves with $\gcd(n, p) = 1$ —the structural setting of ECDSA-type groups, though our experiments are at toy sizes—and hit exactly this case: a higher-order Fourier probe returned a clear $5\text{--}11\sigma$ effect on the original sweep, a textbook control endorsed it, and only a more careful experiment overturned it. The overturning, and the reason the anomaly *had* to be spurious, are the substance of this paper.

Contributions.

- A worked, quantitative example of a *control that does not control*: a permutation test that appears to certify structure but provably cannot separate the competing explanations (§4).
- *Mechanism-derived controls*: enumerate the candidate generating mechanisms and build one null per mechanism, including a *positive* control in which a synthetic model of the surviving mechanism reproduces the statistic quantitatively, a *converse* control—a genuine section deliberately built (by kernel offsets) to break the antisymmetry while preserving everything else, on which the elevation collapses at full sequence length—and a *sham* control (an odd kernel offset) on which it does not (§5). This localizes the anomaly (an elevation of standardized size $\sim 3\text{--}14$ on the clean controlled set) to one exact public identity.
- A closed-form account of the artifact (§5, Prop. 1): a pseudo-variance identity that derives the mass-inflation factor of *any* index-coupling constraint, recovering the reflection’s exact $3/2$ and the $\approx 0.427\sqrt{L}$ growth law as the maximal case, and predicting (via a carry model) the small residual spectral mismatch as a falsifiable effect at $L \gtrsim 3000$.
- An inertness theorem (§7), stated for an arbitrary abelian-group extension, explaining why such anomalies are forced: for $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is $[k]c$, governed by $k \bmod p^{e-1}$, which the public data (E, G, Q) does not determine over the natural lift (and which no known efficient algorithm computes from it—a state-of-knowledge remark, kept separate from the theorem). We present it as the explicit instance it is of [2, 3], with proof and verification.
- A sharp-boundary demonstration (§8): the same construction on an anomalous curve ($\gcd(n, p) = p$) recovers the secret with 100% success (52/52 over all nonzero residues). The theorem’s one hypothesis is exactly the line between inert and broken.
- A scope map (§11): precisely which attack schema this refutes, which it does not, and a named, checklist-form discipline—the *Mechanism-Resolved Null Protocol*—future searches should follow.

The theorem is not new mathematics; its role is to certify, after the fact, that the methodological suspicion was structurally warranted. We stress at the outset that the experiments are diagnostic: they expose and explain a false positive in a lift-error statistic, they are not performed at cryptographic sizes, and they constitute no empirical security evaluation of any deployed ECDSA curve. The extension to ordinary prime-order ECDSA-type curves is carried by the structural theorem, which needs only $\gcd(n, p) = 1$, not by the curve sizes tested here.

2 Background and notation

Let $p \geq 5$ be prime and $E/\mathbb{F}_p$ an elliptic curve with $\#E(\mathbb{F}_p) = N$. Fix $G \in E(\mathbb{F}_p)$ of order n with $\gcd(n, p) = 1$, and $Q = kG$; the ECDLP is to recover $k \in \mathbb{Z}/n\mathbb{Z}$ from (E, G, Q) . For $e \geq 1$, reduction modulo p on a fixed Weierstrass lift gives the exact sequence of abelian groups

$$0 \longrightarrow \widehat{E}(p\mathbb{Z}/p^e) \longrightarrow E(\mathbb{Z}/p^e) \xrightarrow{\text{red}} E(\mathbb{F}_p) \longrightarrow 0, \quad (1)$$

with kernel of reduction $\widehat{E}(p\mathbb{Z}/p^e)$ a p -group of order p^{e-1} [9]. The formal parameter $z = -X/Y$ identifies $\widehat{E}$ with $(p\mathbb{Z}/p^e, +_F)$ ($+_F$ the formal group law), and the formal logarithm $\log: \widehat{E} \rightarrow (p\mathbb{Z}/p^e, +)$ is an isomorphism with $\log([m]u) = m \log(u)$. For $E: y^2 = x^3 + b$ the invariant differential is $\omega = (1 + 3bz^6 + \dots) dz$, so $\log(z) = z + \frac{3b}{7}z^7 + \dots$; for $z \in p\mathbb{Z}$ and $e \leq 4$ the correction vanishes ($z^7 \in p^7\mathbb{Z}$), hence $\log(z) \equiv z \pmod{p^e}$ and kernel arithmetic is additive in z at our precision.

A *section* is a map $\tau: E(\mathbb{F}_p) \rightarrow E(\mathbb{Z}/p^e)$ with $\text{red} \circ \tau = \text{id}$; the Teichmüller and Hensel lifts are sections, generally not homomorphisms. We use the x -Teichmüller / Hensel section throughout; it satisfies $\tau(-P) = -\tau(P)$ (the two Hensel roots in y are negatives), a property we invoke once below. The *lift error* (a 1-cochain valued in $\widehat{E}$ by (1)) is

$$\delta_\tau(k) := [k]\tau(G) - \tau(kG) \in \widehat{E}(p\mathbb{Z}/p^e). \quad (2)$$

We write $\delta(k)$ for the Teichmüller case and $z(\delta(k)) \in p\mathbb{Z}/p^e$ for its formal parameter; for an integer $z \in \mathbb{Z}/p^e$, $\text{digit}_j(z)$ is the coefficient of p^j in the base- p expansion (digit 0 is the units place, identically 0 here since $z \in p\mathbb{Z}$).

Symbol	Meaning
$\delta_\tau(k) = [k]\tau(G) - \tau(kG)$	lift error of section τ (in $\widehat{E}$); δ = Teichmüller case
$z(\delta(k)) \in p\mathbb{Z}/p^e$	formal parameter $-X/Y$ of $\delta(k)$; additive on $\widehat{E}$ at $e \leq 4$
$\text{digit}_j(z)$	coefficient of p^j in the base- p expansion of z
$f_j(k) = e^{2\pi i \text{digit}_j(z(\delta(k)))}/p$	phase encoding of digit j , $k = 1, \dots, n-1$
$\ f\ _{\text{U}^2}^4 = \sum_{\xi} \hat{f}(\xi) ^4$	U^2 <i>mass</i> (mean $2/L$, SD $\Theta(L^{-3/2})$ under i.i.d.)
$\ f\ _{\text{U}^2} = (\text{mass})^{1/4}$	U^2 <i>norm</i> (SD $\Theta(L^{-3/4})$ under i.i.d.)
$C_n = z([n]\tau(G))$	antisymmetry constant of (3); public

Table 1: Notation. The mass/norm distinction matters for standardized effect sizes (Prop. 2).

Prior art. That a p -adic lift transports the ECDLP without weakening it, that the canonical (Teichmüller) lift carries no information, and that “solving the DLP is equivalent to constructing a non-canonical lift” are due to Silverman [2, 1] for elliptic curves and to Gadiyar–Maini–Padma and Gadiyar–Padma [3] for $\mathbb{F}_p^\times$. Smart, Satoh–Araki and Semaev [4, 5, 6] give the anomalous attack; the canonical lift is Lubin–Serre–Tate [7], its computational instance Satoh’s point counting [8].

Yasuda [11] studies the ECDLP transported to the p -adic field and to formal groups, where the formal logarithm makes the kernel DLP easy yet an $\mathbb{F}_p$ instance gains nothing—consistent with, and adjacent to, the inertness mechanism here. Our theorem repackages these in the lift-error language. On the methodology side, the principles we import are textbook in spirit: permutation tests are valid only under exchangeability with respect to the null being tested [12], negative and positive controls diagnose miscalibrated nulls [13], and unacknowledged analytic flexibility manufactures separations [14, 15, 16]; randomness-test batteries in cryptography have their own calibration pitfalls [17]. The contribution is not these principles but their disciplined import into an automated cryptanalytic workflow, with the failure dissected to the constant.

3 The exploration, in brief

The full study ran forty-seven numbered phases against the canonical-lift / formal-group attack surface for $j = 0$ (CM by $\mathbb{Z}[\omega]$) and generic ordinary curves; Appendix A indexes them. Probes included pseudorandomness of δ (DFT flatness, maximal Berlekamp–Massey complexity, uniform p -adic valuation, no polynomial/Mahler/multiplicative fit); a homomorphic section from H^2 vanishing (which exists but, being a homomorphism, annihilates the attack); an explicit Serre–Tate canonical lift validated by $\delta_{\text{can}} \equiv 0$; HNP/lattice (LLL, BKZ) attacks on the cocycle; MOV embedding degree; ordinary/supersingular bridges; genus-2 Jacobian covers; Weil pairings; Iwasawa μ ; and a Frobenius/unit-root analysis. Every one was inert. Two exact relations survived, both *public* (computable from the curve alone) and both giving no advantage: an automorphism-equivariance of the leading δ -digit, and the antisymmetry

$$\delta(k) + \delta(n - k) = [n] \tau(G), \quad (3)$$

which holds for any section with $\tau(-P) = -\tau(P)$ (so that $\tau(kG) + \tau(-kG) = O$), and which we confirmed holds *exactly* in the z -coordinate—e.g. for $p = 97$ ($n = 79$; a curve distinct from the $p = 67$, $n = 79$ entry of Table 2), $z(\delta(k)) + z(\delta(n - k))$ takes a single value across all 39 reflection pairs $(k, n - k)$. We emphasize that (3) encodes nothing beyond the familiar negation symmetry $Q \leftrightarrow -Q$ (i.e. $k \leftrightarrow n - k$): it halves the search space, as every implementor already knows, and confers no further advantage.

4 A false positive: the Gowers anomaly

Encode digit j of the lift error as a phase, $f_j(k) = \exp(2\pi i \text{digit}_j(z(\delta(k)))/p)$ for $k = 1, \dots, n - 1$, and measure the Gowers U^2 norm [10] against an S^1 -uniform random baseline (definitions and counts in §6). Across many primes the U^2 norm was elevated $5\text{--}11\sigma$ on the original exploratory sweep (Phase 37, a mixed bag of curves later found to include supersingular and anomalous cases); on the clean controlled set of §5 the elevation is $\sim 3\text{--}14\sigma$ ($z(\text{real vs rand})$ ranges 3.3 to 13.6). U^3 and higher sat at baseline. The leading Fourier coefficient decayed as $|\hat{f}(\alpha)| \sim Cn^{-\beta}$ with $\beta < 1/2$, on CM, $j=1728$, and non-CM curves alike. An internal report concluded this was a genuine, previously unnoticed degree-1 structural property of the Teichmüller cocycle “not explained by any universal identity.”

The evidence offered was a permutation control: randomly permuting the *values* of f_j destroyed the elevation, read as proof that the structure lived in the index ordering, hence was “real.” This inference is invalid, and seeing why is the point of the paper.

5 Mechanism-derived controls

The principle. A control is a null model, and a null tests exactly one alternative: the one under which the statistic is recomputed. The value-permutation control recomputes U^2 under “the multiset of digit values, in random order,” so it separates “structure in the value distribution” from “structure in the ordering”—and nothing else. But at least three mechanisms predict ordered structure here, and the permutation kills all three at once, so it cannot attribute the signal to any one:

- (H1) *Prefix-sum artifact.* $\delta(k) = -\sum_{i < k} c(iG, G)$ is a running sum; summation is a low-pass filter ($\hat{s}(\xi) = \hat{d}(\xi)/(1 - e^{2\pi i \xi/n})$, a $1/\xi$ pole at $\xi = 0$), so *any* increments with this multiset, once summed, carry elevated low-frequency U^2 .
- (H2) *Antisymmetry shadow.* The exact reflection (3) couples Fourier mode α with $-\alpha$, a rigid constraint that elevates U^2 .
- (H3) *Genuine new structure,* the claimed explanation.

We build one null per mechanism, each derived from how that mechanism generates ordering. All numbers below are on the clean ordinary set of Table 2 (Phase 43), under the pre-specified protocol of §6.

H1 (increment shuffle). $s(k) = z(\delta(k))$ is the exact cumulative sum of its first differences $d(k) = s(k+1) - s(k)$. Shuffle d , re-sum in the new order, recompute U^2 ; the increment multiset is identical, only the order changes. The shuffled-increment prefix sums sit *at* the random baseline ($z(\text{incShuf vs rand}) \in [-0.3, +0.5]$ on every curve), while the true ordering sits +3.5 to +12.5 above them, growing with n (Table 2, cols. 4–5). H1 is refuted: the digit nonlinearity destroys the walk’s low-frequency mass, so the signal is in the *specific* order, not in summation.

H2 (half-sequence). The reflection (3) relates each k to $n - k$. On the first half $k = 1, \dots, \lfloor (n-1)/2 \rfloor$ every reflection partner lies outside the window, so H2 predicts no internal constraint there while H3 predicts the signal persists. Re-running the increment-shuffle control on the half collapses the elevation to noise ($z(\text{half vs incShuf}) \in [-1.3, +1.5]$, no growth in n ; Table 2, col. 6).

H2, positive control (the decisive test). Absence of signal on the half is necessary but not sufficient. We therefore test H2 *constructively*: generate synthetic sequences that impose *only* the reflection, $s(k)$ i.i.d. uniform on the first half and $s(n-k) = C - s(k)$ with $C = z([n]\tau(G))$, carrying no other structure, and measure their U^2 . If reflection alone is the mechanism, these must reproduce the real statistic. They do, quantitatively: the synthetic U^2 is elevated +3.8 to +12.5 over the random baseline—tracking the real elevation across the whole range of n —and the real sequences are statistically indistinguishable from them ($z(\text{real vs syn}) \in [-1.4, +1.5]$; Table 2, cols. 7–8). The match on the U^2 elevation and its growth with n is the substance of the positive control, and it is exact (cols. 7–8). The finer digit-dependent Fourier decay is only *partially* reproduced: fitting $|\hat{f}| \sim Cn^{-\beta}$ across the seven curves, the real sequences give $\beta_{\text{real}} \approx \{0.34, 0.45, 0.36\}$ for digits 1, 2, 3, whereas the reflection-only synthetic draws give a nearly flat $\beta_{\text{syn}} \approx \{0.41, 0.41, 0.40\}$. Reflection alone thus reproduces the overall power-law decay ($\beta \approx 0.4$, the same order as the real exponents) but not its digit-to-digit variation; we do *not* claim the spectra match in detail. The residual digit-to-digit variation is, however, consistent with finite-size fit scatter rather than secondary structure: resampling the seven-curve fit on reflection-only sequences gives β_{syn} with

mean ≈ 0.40 and standard deviation ≈ 0.06 *with no digit dependence built in*, and each real exponent $\{0.34, 0.45, 0.36\}$ lies within ≈ 1 standard deviation of that mean. The mild downward spread of the interior-digit exponents is moreover the *sign* predicted by the carry model of §5 (Prop. 2): digit 1 reflects exactly while digits 2, 3 carry a small borrow-induced deficit. Under the standard assumption that the ECDLP is hard on these curves, the residual cannot be cryptanalytic in any case—it is a deterministic post-processing of the public digit map of a public object (Theorem 1, Corollary 1 and the Consequence following it)—so we do not pursue it further; the decisive evidence that H2 is the mechanism is the U^2 match, which is genuine. A post-hoc confirmatory run on a large-prime-order curve ($p = 10477$, $n = 10639$ prime; Phase 47, last row of Table 2) extends the pattern by more than an order of magnitude in n : the real elevation reaches $+43$ – 44 , the reflection-only synthetics track it at $+44.5$ – 44.8 , and real vs. synthetic stays within ± 2.1 .

The converse control (breaking the reflection). A positive control can be accused of circularity—a synthetic sequence built with the reflection will of course exhibit the reflection statistic. The decisive converse is to remove the reflection from the *real* construction and check that the elevation disappears. The antisymmetry (3) holds only because the section is odd, $\tau(-P) = -\tau(P)$; the converse must therefore break oddness *while remaining a genuine section*, so that nothing else changes. (A first attempt, Phase 49, chose the Hensel y -root by a negation-insensitive “smaller integer representative” rule; that rule is *not* a section—it lifts $-P$ rather than P on roughly half the points, so half the lift errors leave the kernel of reduction and the degeneracy filter halves the usable length. The collapse it showed was therefore confounded, as Phase 49b documented; we retain it only as a cautionary record.) The clean construction perturbs the odd section by a *kernel offset*:

$$\tau'(P) := \tau(P) + [h(P)] K_1, \quad K_1 := [n]\tau(G) \in \widehat{E}, \quad h: E(\mathbb{F}_p) \rightarrow \mathbb{Z}/p^{e-1},$$

where h is a deterministic function of the point. Because the offset lies in the kernel, $\text{red} \circ \tau' = \text{id}$ for *every* P : τ' is a genuine section, $\delta_{\tau'}(k)$ stays in the kernel for all k , and the usable length is the full $n - 1$ with zero drops. Oddness fails exactly when h is not odd, and in z -coordinates the new lift error is computable in closed form, $z(\delta'(k)) = z(\delta(k)) + (k h(G) - h(kG)) C_n \bmod p^e$. We ran four representative curves, $p \in \{67, 211, 823, 10477\}$ (Phase 60, same secrets and encoding as Table 2). With a *generic* non-odd offset—one whose values fill the kernel, $h(P) = x(P)$ or $h(P) = x(P) + 7y(P) \bmod p^{e-1}$ —the antisymmetry sum $z(\delta'(k)) + z(\delta'(n - k))$ ceases to be single-valued ($\lfloor (n-1)/2 \rfloor$ distinct values, one per pair) and the U^2 elevation collapses from $z(\text{real vs rand})$ of $+3.6, +6.6, +11.8, +45.9$ (digit-1 values across the four curves) to baseline ($z \in [-1.6, +1.9]$ across all curves and digits). With an *odd* offset ($h(P) = y(P)$, a sham control for which τ' remains odd) the identity stays single-valued and the elevation remains ($+3.4, +4.7, +9.5, +47.8$ at digit 1). Breaking oddness with a generic offset destroys identity and elevation together; preserving oddness under an equally disruptive kernel perturbation preserves both.

One caveat keeps the claim honest, and it is exactly what Prop. 1 predicts. Destroying the reflection’s single-valuedness is *necessary* but not *sufficient*: a non-odd offset that is itself *low-rank* can substitute one index constraint for another and so keep U^2 elevated for a different reason. We see this with a deliberately degenerate binary offset ($h(P) \in \{0, 1\}$ by a half-plane rule): single-valuedness is lost, yet U^2 stays elevated (e.g. $+38, +38, +19$ at $p = 10477$), because the two-valued offset imposes its own low-pseudo-variance constraint. The correct reading is therefore the pseudo-variance one: what collapses the elevation is removing the per-mode conjugacy *without introducing a new low-rank index constraint*, which the generic full-entropy offsets x and $x + 7y$ do and the binary offset does not. The mechanism is the reflection—read off the constraint’s action on Fourier indices, not off single-valuedness alone.

The growth law is itself an artifact signature. The elevation does not merely persist with n ; it *grows*, from ~ 3 at $n = 79$ to ~ 44 at $n = 10639$. It is tempting to read growth-with- n as accumulating signal—this is precisely what the internal Phase-37 report did. The opposite is true: the growth is forced by the reflection alone and carries no curve information, as the following closed form shows.

We treat the general mechanism first and read off the reflection as the extremal case. Write the U^2 mass as $\|f\|_{U^2}^4 = \sum_{\xi} |\hat{f}(\xi)|^4$ with $\hat{f}(\xi) = \frac{1}{L} \sum_k f(k) e^{-2\pi i k \xi / L}$. In the large- L limit each nonzero mode $Z_{\xi} := \hat{f}(\xi)$ is a mean-zero complex Gaussian characterized by two second moments: the *variance* $\sigma^2 = \mathbb{E}|Z_{\xi}|^2 = 1/L$ and the *pseudo-variance* $\rho(\xi) = \mathbb{E}[Z_{\xi}^2]$, the latter zero for a proper (circularly symmetric) mode and nonzero exactly when a real-linear constraint ties Z_{ξ} to its conjugate $\overline{Z_{\xi}}$ (equivalently, confines the mode to a line in $\mathbb{C}$). The distinction is essential and is the point a conjugation makes: a value-domain reflection *with* conjugation produces this; the same reflection *without* it does not (Phase 63, and the proof below).

Proposition 1 (Pseudo-variance classification of U^2 inflation). *For a complex Gaussian Z with variance σ^2 and pseudo-variance ρ ,*

$$\mathbb{E}|Z|^4 = 2\sigma^4 + |\rho|^2. \quad (4)$$

Hence for $f: \mathbb{Z}/L \rightarrow S^1$ whose modes have per-mode pseudo-variance $\rho(\xi)$, $\mathbb{E}[\|f\|_{U^2}^4] = (2 + \langle |\rho|^2 \rangle / \sigma^4) / L + o(1/L)$, where $\langle \cdot \rangle$ averages over the $L - 1$ nonzero modes, and the mass-inflation factor relative to the i.i.d. baseline is

$$INFLATION = 1 + \frac{\langle |\rho(\xi)|^2 \rangle}{2\sigma^4} \in [1, \frac{3}{2}]. \quad (5)$$

The upper bound $|\rho| \leq \sigma^2$ (Cauchy–Schwarz) is attained, giving the maximal $3/2$, iff $|\rho(\xi)| = \sigma^2$ a.e.—the case of a per-mode conjugacy constraint $\hat{f}(\xi) = \beta_{\xi} \overline{\hat{f}(\xi)}$ ($\beta_{\xi} \in S^1$), each mode confined to a line, which is the Fourier image of a value-domain conjugate reflection $f(c - k) = \beta \overline{f(k)}$. A constraint that merely fixes or permutes modes without tying them to their conjugates (e.g. a translation $k \mapsto k + s$, or the non-conjugate reflection $f(c - k) = \beta f(k)$, which couples ξ to $-\xi$ but leaves each mode proper) gives $\rho \equiv 0$ and no inflation.

Proof. Write $Z = X + iY$ with real covariance $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$, so $\sigma^2 = a + b$ and $\rho = (a - b) + 2ic$, $|\rho|^2 = (a - b)^2 + 4c^2$. Isserlis’ theorem gives $\mathbb{E}[X^4] = 3a^2$, $\mathbb{E}[Y^4] = 3b^2$, $\mathbb{E}[X^2Y^2] = ab + 2c^2$, whence $\mathbb{E}|Z|^4 = \mathbb{E}[(X^2 + Y^2)^2] = 3a^2 + 3b^2 + 2ab + 4c^2 = 2(a + b)^2 + (a - b)^2 + 4c^2 = 2\sigma^4 + |\rho|^2$, proving (4). Summing over modes and using $\sigma^2 = 1/L$ gives the mass and (5); the bound $|\rho| \leq \sigma^2$ is Cauchy–Schwarz on $\mathbb{E}[Z^2]$ versus $\mathbb{E}|Z|^2$. A value-domain conjugate reflection $f(c - k) = \beta \overline{f(k)}$ forces, per mode, $Z_{\xi} = \beta \omega^{-c\xi} \overline{Z_{\xi}}$ ($\omega = e^{2\pi i/L}$), confining Z_{ξ} to a line; hence $\mathbb{E}[Z_{\xi}^2]$ has modulus $\mathbb{E}|Z_{\xi}|^2 = \sigma^2$. Without the conjugation the constraint $f(c - k) = \beta f(k)$ instead forces $Z_{\xi} = \beta \omega^{-c\xi} Z_{-\xi}$, which couples ξ to $-\xi$ but leaves each mode proper, so $\mathbb{E}[Z_{\xi}^2] = 0$ and there is no inflation. We confirmed both numerically: the per-mode ratio $\langle |\rho(\xi)| \rangle / \sigma^2$ is 1.00 for the conjugate reflection and 0.01 for the non-conjugate one and for i.i.d. phases (Phase 63). $\square$

Proposition 2 (The reflection inflates the U^2 mass by a limiting factor of exactly $3/2$). *Let $f: \mathbb{Z}/L \rightarrow S^1$ be subject to only the antisymmetry constraint $f(L+1-k) = \beta \overline{f(k)}$ ($\beta \in S^1$ fixed, the free half i.i.d. uniform)—the reflection-only synthetic of the positive control. (The conjugation is the phase image of the value identity $s(n-k) = C - s(k)$, since $e^{2\pi i(C-s)/p} = e^{2\pi i C/p} \overline{e^{2\pi i s/p}}$; it is essential—the non-conjugate $f(L+1-k) = \beta f(k)$ gives no inflation.) Then $\mathbb{E}[\|f\|_{U^2}^4] = 3/L +$*

$o(1/L)$, versus $2/L + o(1/L)$ for i.i.d. uniform entries; the limiting mass ratio is exactly $3/2$, independent of the value alphabet (continuous S^1 or uniform p -th roots of unity alike; Phase 63). The standardized synthetic-vs-random gap, measured on the U^2 norm against the i.i.d. null, is $z(\text{syn/rand}) = 2^{7/4}(3^{1/4} - 2^{1/4})\sqrt{L} + o(\sqrt{L}) \approx 0.427\sqrt{L}$.

Proof. Mean. The conjugate reflection confines every mode to a line (per-mode conjugacy), so by Prop. 1 the inflation factor is $3/2$, i.e. $\mathbb{E}[\|f\|_{U^2}^4] = 3/L + o(1/L)$. The finitely many exceptional modes (the zero mode, whose magnitude is $O_P(L^{-1})$ in mass either way; the self-paired mode $\xi = L/2$ for even L ; the fixed physical midpoint for odd L) each contribute $O(1/L^2)$ to the mass and are absorbed in the $o(1/L)$.

Variance of the i.i.d. mass—the step that fixes the constant. Write $M = \sum_{\xi} p_{\xi}^2$ with $p_{\xi} = |\hat{f}(\xi)|^2$. Parseval forces $\sum_{\xi} p_{\xi} = \frac{1}{L} \sum_k |f(k)|^2 = 1$ exactly, so the periodogram is a random point on the simplex; for random phases it is, in the large- L limit, well approximated by a uniform Dirichlet($1, \dots, 1$) vector on L parts—not a vector of independent exponentials, the approximation we make and then confirm numerically. Under it, the Dirichlet moment formula $\mathbb{E}[\prod_i p_i^{a_i}] = \prod_i a_i! \cdot \Gamma(L)/\Gamma(L + \sum a_i)$ gives

$$\mathbb{E}[M] = \frac{2}{L+1}, \quad \mathbb{E}[M^2] = \frac{4(L+5)}{(L+1)(L+2)(L+3)}, \quad \text{Var}(M) = \frac{4}{L^3} + O(L^{-4}).$$

Ignoring the Parseval constraint would give $\text{Var}(M) = L \cdot \text{Var}(|Z|^4) = 20/L^3$ (since $\text{Var}(|Z|^4) = 20\sigma^8$ for a proper complex Gaussian) and hence a growth constant ≈ 0.19 , contradicting experiment; the constraint reduces the variance by exactly the factor 5. So $\text{SD}(M) = 2L^{-3/2} + O(L^{-2})$, and by the delta method on the *norm* $M^{1/4}$ (mean $(2/L)^{1/4}$), $\text{SD}(\text{norm}) = \frac{1}{4}(2/L)^{-3/4} \cdot 2L^{-3/2} = 2^{-7/4}L^{-3/4}$.

Standardized gap. The mean gap in the norm is $(3/L)^{1/4} - (2/L)^{1/4} = (3^{1/4} - 2^{1/4})L^{-1/4}$; dividing by $2^{-7/4}L^{-3/4}$ gives $2^{7/4}(3^{1/4} - 2^{1/4})\sqrt{L}$. $\square$

The exact Dirichlet computation and a Monte Carlo confirmation that $\text{Var}(M)$ tracks $4/L^3$ (and not $20/L^3$) across $L \in [78, 3000]$, together with the delta-method check $\text{SD}(\text{norm})/2^{-7/4}L^{-3/4} \in [0.96, 1.01]$, are Phase 61.

The coefficient $2^{7/4}(3^{1/4} - 2^{1/4}) = 0.4267\dots$ is fully analytic; it is the closed form of the constant the original sweep had only measured. Standardizing the U^2 norm against the empirical-null standard deviation, we confirmed it p - and L -independent across $L \in [78, 10^6]$ (Phases 48, 56): the mass ratio holds to 1.500 at $L = 10^6$, and $0.427\sqrt{n}$ reproduces the syn/rand column of Table 2 and the Phase-47 value (predicted 44.8 vs. observed 44.6 at $n = 10639$) within finite-size noise. So $\sqrt{n}$ growth is the *signature of the antisymmetry artifact*, not of an accumulating signal—the exact inference the original report inverted. Equation (5) also makes the discriminator quantitative: a constraint sending mode ξ to its reversal $-\xi$ inflates the mass to $3/L$, whereas one sending it elsewhere (e.g. the fixed-shift involution $k \mapsto k + L/2$) leaves it at $2/L$; we verified both numerically (ratios 1.499 vs. 1.001; Phase 55), so an auditor reads the mechanism off the constraint’s action on Fourier indices. Figure 1 summarizes the evidence: the separated null distributions, the $0.427\sqrt{L}$ law against the data, and the hypothesis-by-control matrix whose first row is the permutation control’s failure.

Higher norms, and a carry prediction. Two consequences of the same analysis sharpen the picture. First, the inflation is a purely second-order (quadratic-mass) phenomenon: heuristically, the extra difference operator defining U^3 couples reflection partners only on an $O(1/L)$ fraction of shift–position pairs, so $\mathbb{E}\|f\|_{U^3}^8$ should sit at the unconstrained baseline up to $O(1/L)$. We state this

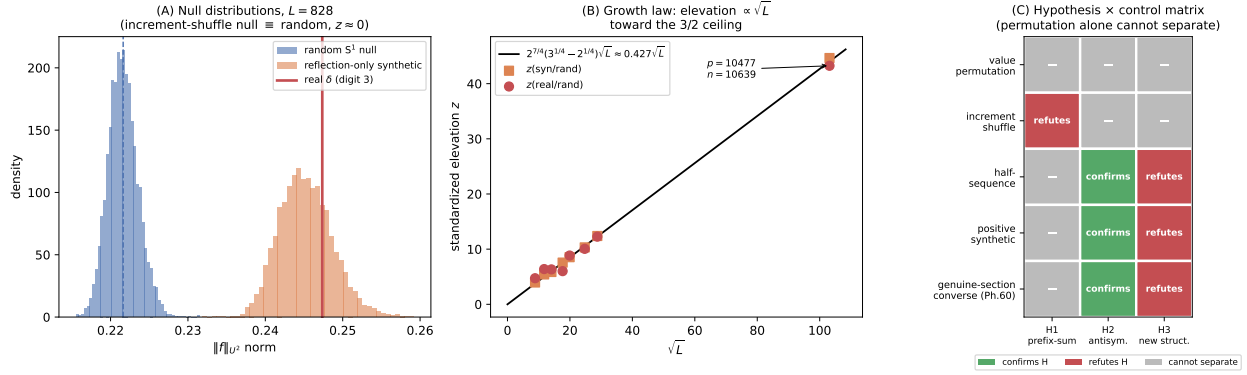


Figure 1: (A) Null distributions of the U^2 norm at $L = 828$: the S^1 -random baseline, the reflection-only synthetic, and the measured real value (digit 3); the increment-shuffle null coincides with the random null ($z \approx 0$ on every curve). (B) Standardized elevation versus $\sqrt{L}$ for the seven clean curves and the confirmatory $p = 10477$: both $z(\text{syn/rand})$ and $z(\text{real/rand})$ follow the closed-form $2^{7/4}(3^{1/4} - 2^{1/4})\sqrt{L} \approx 0.427\sqrt{L}$ line of Prop. 2. (C) Hypothesis $\times$ control matrix: the value-permutation control kills all three mechanisms at once and so separates none; each mechanism-derived control implicates or refutes a named hypothesis, and the genuine-section converse (Phase 60) completes the attribution. Generated by Phase 62 from committed results.

as a counting heuristic rather than a proof, and confirm it numerically: the U^3 excess decays as $1/L$ while the U^2 ratio holds at $3/2$ (Phase 59). This accounts for the otherwise unexplained observation that U^3 and higher sat at baseline. Second, the real digit sequence reflects only approximately: base- p borrows mean the phase-level reflection $\text{digit}_j(C-s) \leftrightarrow \text{digit}_j(s)$ holds for a fraction $1 - q_j$ of pairs. A simple independence model for the borrow failures—a modelling assumption, not a derived fact—predicts per-mode $|\rho| = (1 - q_j)\sigma^2$, hence by (5) a mass $(2 + (1 - q_j)^2)/L$. Digit 1 reflects *exactly* ($q_1 = 0$, since $z(\delta) \in p\mathbb{Z}$ forces digit 0 to vanish on both sides, so no borrow reaches position 1); interior digits have $q_2, q_3 \approx 0.2$ measured on the clean set—a value that depends on the constant C , the digit distribution, p , and the sample, and is not claimed to be universal—predicting a mild mass deficit there. At the clean-set sizes this deficit (~ 0.3 in L -mass) sits at the $\lesssim 2\sigma$ resolution limit and is *not* yet distinguishable from finite-size scatter—consistent with the small β_{real} spread we report below—so we state it as a falsifiable prediction: a confirmatory run at $L \gtrsim 3000$ should resolve U^2 masses of ≈ 3.0 at digit 1 and ≈ 2.7 at digits 2, 3 (Phase 58).

Verdict. The anomaly is the Fourier-domain shadow of the exact, public identity (3): H2, not H1 or H3. It is real but not new, and—being a public consequence of the curve’s geometry that yields only a 2-to-1 search reduction—carries no cryptanalytic advantage. The varying “decay exponent” and its apparent curve-family dependence in the internal report are finite-size scatter around a single reflection-induced spectrum; nothing family-specific survives. To rule out a complex-multiplication artifact—the clean set is $j=0$, which carries extra automorphisms—we re-ran the full protocol on $j=1728$ and on generic non-CM curves ($j \in \{6, 316, 446\}$, prime order, $\gcd(n, p) = 1$): the $3/2$ inflation, the syn/rand tracking, and the real-vs-syn equivalence reproduce identically, confirming the mechanism is the antisymmetry and not the CM structure (Phase 50; per-curve numbers in Table 6, Appendix B).

p	n	L	negative controls (z)			positive (z)	
			real/inc	inc/rand	half/inc	syn/rand	real/syn
67	79	78	+3.5..6.7	−0.3..0.2	−0.6..1.5	+3.8..4.1	−0.4..1.2
163	139	138	+3.7..7.4	+0.0..0.5	−0.4..1.0	+5.4..5.4	−0.9..1.5
211	199	198	+5.1..7.8	+0.1..0.3	−0.8..0.5	+5.8..5.9	−0.3..0.8
349	313	312	+5.4..7.1	−0.0..0.1	−1.3..0.2	+7.5..7.6	−1.4.. −0.4
433	397	396	+8.2..9.6	−0.1..0.2	−0.4..0.6	+8.4..8.6	−0.1..0.7
577	613	612	+7.8..11.5	+0.1..0.2	−0.5..1.1	+10.2..10.4	−1.2..0.8
823	829	828	+10.5..12.5	+0.1..0.2	−0.9..0.0	+12.2..12.5	−0.9..0.7
10477 [†]	10639	10638	+42.7..44.1	−0.0..0.0	−1.1..1.8	+44.5..44.8	−2.0..0.7

Table 2: Phase 43, clean ordinary $j=0$ curves $y^2 = x^3 + 7$ ($p \equiv 1 \pmod{3}$, $n = \#E(\mathbb{F}_p)$ prime, so $\gcd(n, p) = 1$ and the curve is neither supersingular nor anomalous); L is the number of usable $k \in [1, n - 1]$ after the pre-specified degeneracy filter (§6); here $L = n - 1$ (no drops). Ranges are over digits $j = 1, 2, 3$. Negative controls: shuffling increments leaves U^2 at the random baseline (inc/rand) and the true ordering far above it (real/inc), which vanishes on the half (half/inc). Positive control: synthetic reflection-only sequences reproduce the elevation (syn/rand) and the real sequences match them (real/syn). The entire effect is the antisymmetry (3). The last row ([†], Phase 47, $p = 10477$) is a post-hoc confirmatory run under the identical protocol, reported separately from the pre-specified seven-curve set.

6 Statistical protocol

All quantities in §5 follow one fixed analysis plan (pre-specified before the clean re-run as an immutable seed and protocol in the repository, seed 20260609; we call it “fixed” rather than “pre-registered” as there is no external timestamped registration). Precision $e = 4$; digits $j \in \{1, 2, 3\}$. The S^1 -random baseline draws 300 sequences $g(k) = e^{i\theta_k}$, θ_k uniform on $[0, 2\pi)$, of the same length L ; “ σ ” and “ z ” are the standard deviation and standardized deviation *of the relevant null*: incShuf uses 300 increment shuffles, syn uses 300 synthetic draws, rand uses the 300 baselines. The Gowers U^2 is computed exactly by its FFT formula. *Exclusion rule (pre-specified)*: a value $z(\delta(k))$ is dropped iff its projective lift is degenerate— Y a non-unit, or $z \not\equiv 0 \pmod{p}$ (a kernel element must have $z \equiv 0$); these are arithmetic failures of the pure-Python lift, flagged automatically, and on the clean ordinary set none occurred ($L = n - 1$ throughout). The clean set is the seven ordinary $j=0$ curves $y^2 = x^3 + 7$ with $p \equiv 1 \pmod{3}$ and $n = \#E(\mathbb{F}_p)$ prime (hence $\gcd(n, p) = 1$, neither supersingular nor anomalous); the smallest has $n = 79$, so $L \geq 78$ throughout. No further size-based filter is applied—every such curve in the run is reported. We make no multiplicity correction and therefore do *not* claim discovery from any single elevated U^2 ; the paper’s logic is the opposite—every elevation is traced to a named mechanism and shown to be identity-or-artifact, and the controls are designed to *remove* signal, not to find it. Phase 47 repeats this protocol verbatim on one large-prime-order curve ($p = 10477$, $n = \#E(\mathbb{F}_p) = 10639$ prime); being post-hoc, it is reported separately (marked [†] in Table 2). Throughout, z -values are standardized effect sizes against 300-draw empirical nulls, not calibrated Gaussian tail probabilities; we use them only comparatively, never as discovery statistics.

Calibration. Three checks make the evidential language precise. (i) *Look-elsewhere*. The exploration spans the original 47-phase sweep $\times$ 3 digits $\times$ 8 curves $\times$ several encodings (the later phases 48–63 are controls and calibration, not search); the expected maximum $|z|$ of a generic

random baseline over a battery of this size is ≈ 3.2 – 3.6 by extreme-value theory, and an empirical 10-statistic battery on i.i.d. surrogates reached $|z| = 5.0$ (Phase 54). A “ 5σ ” separation from a generic baseline is therefore *expected* under this much exploration. But look-elsewhere arithmetic explains only the moderate outliers; it cannot manufacture the 11–44 range. Those are *wrong-null* false positives: a real, public algebraic identity tested against a generic random baseline produces a separation that grows without bound ($\propto \sqrt{L}$, Prop. 2) while carrying no secret information. The two mechanisms require different defenses—multiplicity control for the first, mechanism-derived nulls for the second—and conflating them is itself a failure mode. (ii) *Equivalence, not absence of significance*. For the real-vs-synthetic comparison we report a two-one-sided-test bound, specified as follows: the statistic is the U^2 norm; the null is the 300-draw reflection-synthetic distribution (mean μ_{syn} , SD s_{syn}); each digit is tested separately; and the excluded residual is the 95% one-sided upper confidence limit $\Delta_{\text{excl}} = (\max(0, z_{\text{obs}}) + 1.645) s_{\text{syn}}$ on any non-reflection (“H3”) component beyond the synthetic baseline, reported as a fraction of the elevation $E = U_{\text{real}}^2 - \mu_{\text{rand}}$. This fraction tightens from ≈ 0.9 at the smallest curve to ≈ 0.25 at $p = 823$ (Phase 52). Equivalence is thus established tightly only at larger L ; we say “indistinguishable” throughout in exactly this bounded sense. (iii) *Power of the half-sequence control*. Because the gap grows as $\sqrt{L}$, an intrinsic (length-extensive) H3 signal of the observed full-length size would still appear at $\approx z_{\text{full}}/\sqrt{2}$ on the half. The minimum detectable effect at $\alpha = 0.05$ (two-sided) and power 0.80 is $z = 2.80$ against the half-length increment-shuffle null; the per-curve targets $z_{\text{full}}/\sqrt{2}$ range from 2.5 to 8.8 while the observed half-sequence values lie in $[-1.3, +1.5]$ with no growth in n (Table 2, col. 6). This excludes an intrinsic H3 on 6 of the 7 curves; only the smallest ($p = 67$, where two of three digit targets fall below the MDE) is genuinely underpowered (Phase 53).

7 Why it had to be a false positive: an inertness theorem

For $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is forced into a coordinate the attacker cannot compute, so any “structure” in δ that *is* computable from (E, G, Q) must reduce to a public identity or an artifact. We make this precise.

Lemma 1 (Canonical lift). *If $\gcd(n, p) = 1$, the order- n subgroup $\langle G \rangle$ lifts uniquely to $\langle \tilde{G} \rangle \subset E(\mathbb{Z}/p^e)$ of order n with $\text{red}(\tilde{G}) = G$, and $jG \mapsto [j]\tilde{G}$ is a homomorphism (so $\delta_{\tau_{\text{can}}} \equiv 0$). Explicitly $\tilde{G} = [m]\tau(G)$ for any section τ , where $m \equiv 1 \pmod{n}$, $m \equiv 0 \pmod{p^{e-1}}$.*

Proof. $E(\mathbb{Z}/p^e)[n]$ injects into $E(\mathbb{F}_p)[n]$ (an n -torsion element of $\hat{E}$ is killed by $\gcd(n, p^{e-1}) = 1$, hence trivial) and surjects onto it (étale n -torsion lifts by Hensel), so reduction is an isomorphism on n -torsion and $\tilde{G}$ is the unique order- n lift of G . Writing $\tau(G) = \tilde{G} + c$ with $c \in \hat{E}$, $[m]\tau(G) = [m \bmod n]\tilde{G} + [m \bmod p^{e-1}]c = \tilde{G}$. $\square$

Theorem 1 (Inertness). *Let τ be any section and $c := \tau(G) - \tilde{G} \in \hat{E}$. Then:*

- (a) $\delta_\tau(k) = [k]c - c_k$, where $c_k := \tau(kG) - [k]\tilde{G} \in \hat{E}$ depends only on the point kG .
- (b) $[k]c$ depends on k only through $k \bmod p^{e-1}$; $\log([k]c) = k \log(c)$ has additive order a power of p dividing p^{e-1} .
- (c) $\gcd(n, p^{e-1}) = 1$, so by CRT, for k uniform on $\mathbb{Z}/np^{e-1}\mathbb{Z}$ the residues $k \bmod n$ (fixed by Q) and $k \bmod p^{e-1}$ (governing $[k]c$) are independent. The public data (E, G, Q) fixes only $k \bmod n$; hence $[k]c$ is not a section-invariant function of the public instance—it varies with the choice of lift through $k \bmod p^{e-1}$, which (E, G, Q) does not pin down. (This is a section-invariance / well-definedness statement; the stronger computational claim is deferred to the Remark and the Consequence following Corollary 1.)

(d) δ_τ is a well-defined function of the instance (i.e. of $k \bmod n$) iff $c = 0$ (i.e. τ canonical on $\langle G \rangle$), in which case $\delta_\tau \equiv 0$. For $c \neq 0$, $\delta_\tau(k + n) - \delta_\tau(k) = [n]c = [n]\tau(G) \neq O$, the antisymmetry constant of (3).

Proof. (a) $[k](\tilde{G} + c) - ([k]\tilde{G} + c_k) = [k]c - c_k$. (b) $c \in \hat{E}$ has order dividing p^{e-1} ; apply $\log$, whose order as a map $k \mapsto k \log(c)$ on $(p\mathbb{Z}/p^e, +)$ is $p^e / \gcd(\log(c), p^e)$, a power of p dividing p^{e-1} since $\log(c) \in p\mathbb{Z}$. (c) Since $\gcd(n, p^{e-1}) = 1$, CRT gives $\mathbb{Z}/np^{e-1}\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/p^{e-1}\mathbb{Z}$, so for k uniform on the left the residues $k \bmod n$ and $k \bmod p^{e-1}$ are independent; as (E, G, Q) determines only $k \bmod n$, it leaves $k \bmod p^{e-1}$ (and hence $[k]c$) information-theoretically unconstrained over this lift. We make no claim of a reduction beyond this. (d) c_k is a function of kG , hence of $k \bmod n$; so δ_τ factors through $k \bmod n$ iff $k \mapsto [k]c$ does, iff $[n]c = O$, iff (as $\gcd(n, p^{e-1}) = 1$) $c = O$. The shift equals $[n]c = [n]\tau(G)$ since $[n]\tilde{G} = O$. $\square$

Corollary 1. For $\gcd(n, p) = 1$, the lift error δ_τ factors through the instance (i.e. through $k \bmod n$) if and only if $c = 0$, in which case it is identically zero. For $c \neq 0$ it is not a function of the instance at all: it carries the ambiguity $[n]\tau(G) \neq O$, whose only instance-independent shadow is the public antisymmetry constant of (3).

Consequence (under the standard ECDLP-hardness assumption). Combining Corollary 1 with the state-of-knowledge remark below: no section lift error yields an *efficiently computable* function of (E, G, Q) that depends on the secret—exploiting the non-canonical case would require producing the residue information $k \bmod p^{e-1}$ that the public instance does not determine, for which no efficient method is known short of solving the ECDLP. The p -adic canonical lift moves the ECDLP without weakening it. We label this a consequence rather than a corollary because it leans on a computational assumption, not on the group theory alone.

Generality. The proof uses nothing about elliptic curves beyond the exact sequence (1): an extension $0 \rightarrow K \rightarrow B \rightarrow G \rightarrow 0$ of abelian groups with G finite of order n , K a p -group of exponent dividing p^{e-1} , and $\gcd(n, p) = 1$. Then n is invertible on K , so $H^2(G, K) = 0$, the retraction $b \mapsto \mu([n]b)$ with $\mu = ([n]|_K)^{-1}$ splits the sequence, and the lift error is a coboundary carrying no extension data. Theorem 1 is the instance $B = E(\mathbb{Z}/p^e)$, $G = E(\mathbb{F}_p)$, $K = \hat{E}(p\mathbb{Z}/p^e)$; because the hypotheses are purely group-theoretic it ports verbatim to Jacobians of higher-genus curves, algebraic tori, and any commutative group scheme with smooth reduction and fibre order coprime to p . This is what we mean by “regime-agnostic”: the conclusion is a one-line consequence of coprime-order cohomology vanishing, independent of the curve sizes tested.

This is why the Gowers anomaly was forced to be artifact-or-identity. The only secret-bearing object, $[k]c$, is not publicly computable; the visible δ , taken as a function of the integer k , is either zero or periodic-with-shift $[n]\tau(G)$ —and that shift is the reflection (3) whose Fourier signature the probe rediscovered.

Remark (on clause (c)). We deliberately do *not* claim $[k]c$ is information-theoretically independent of (E, G, Q) : once a canonical representative $k \in [1, n - 1]$ is fixed, $k \bmod p^{e-1}$ and hence $[k]c$ are determined. The claim is computational—no efficient algorithm produces $[k]c$ from the public data short of solving the ECDLP—which is what matters for security and what the experiments probe. Being a state-of-knowledge assertion, this claim is confined to the present remark rather than the theorem statement.

Regime	curves	$[n]$ on $\widehat{E}$	invariant $[n]\tau(G)$	ECDLP
Small ordinary, $\gcd(n, p) = 1$	$p = 31\text{--}163, j=0$	automorphism	$\neq O$, ord $= p^{e-1}$, public	inert
Anomalous (Smart), $n = p$	$y^2 = x^3 + 4x + 7/\mathbb{F}_{53}$	zero map	$\neq O$, section-indep., $\propto k$	broken, 52/52

Table 3: Phases 41–42. What vanishes at the boundary is the *ambiguity*, not the invariant. For $\gcd(n, p) = 1$, $[n]$ is an automorphism of the kernel: the well-definedness obstruction of Theorem 1(d) has maximal order and the only well-defined quantity is the public antisymmetry constant of (3). For $n = p$, $[n]$ annihilates the kernel: the section ambiguity $[p]c$ dies and $[p]\tau(G)$ becomes a well-defined, *nonzero*, secret-linear invariant—which Smart’s attack requires ($\varphi(G) \neq 0$). $\gcd(n, p) = 1$ is the exact boundary. Theorem 1 is regime-agnostic—it holds for any $\gcd(n, p) = 1$ —so these small ordinary curves (chosen for exact-arithmetic reach) suffice to verify it; the relevance to large-prime ECDSA curves is through the theorem’s generality, not through these particular orders.

8 The boundary is sharp

Theorem 1 hinges on $\gcd(n, p) = 1$, and what changes at the boundary is the action of $[n]$ on the kernel of reduction. Set $n = p$ (an anomalous curve, $\#E(\mathbb{F}_p) = p$; take $e = 2$): $[n] = [p]$ now *annihilates* the kernel— $[p]c = O$ for every $c \in \widehat{E}(p\mathbb{Z}/p^2)$, since $\log([p]c) = p \log(c) \in p^2\mathbb{Z}$ —whereas for $\gcd(n, p) = 1$ it acts as an automorphism. Two consequences follow. First, the *section ambiguity* dies: two sections differ pointwise by a kernel element, which $[p]$ kills, so $[p]\tau(P)$ is independent of the choice of section. Second, what survives is *not* zero: $\varphi(P) := \frac{1}{p} \log([p]\tau(P)) \bmod p$ is a homomorphism $E(\mathbb{F}_p) \rightarrow \mathbb{F}_p$, linear in the secret, and Smart’s attack is its evaluation, $k = \varphi(Q)/\varphi(G)$. The asymmetry with the ordinary case is exact: there the ambiguity has maximal order and the only well-defined quantity is the public constant of (3); here the ambiguity vanishes and the surviving invariant carries k . (Note that δ itself remains ill-defined as a function of $k \bmod p$: $\delta(k+p) - \delta(k) = [p]\tau(G) \neq O$. The object that becomes well-defined is $[p]\tau(P)$, not δ .) We verified both sides (Table 3): on every ordinary test curve the obstruction $[n]\tau(G)$ is a nonzero kernel element of *maximal* order p^{e-1} (valuation 1), while on the anomalous $y^2 = x^3 + 4x + 7$ over $\mathbb{F}_{53}$ (with $G = (0, 22)$) the value $z([p]\tau(G)) = 15 \cdot 53 \bmod 53^2$ is *section-independent* (identical under the Teichmüller- x , naive- x , and shifted- x sections) and nonzero—as it must be, since the attack divides by $\varphi(G) = 15$ —and Smart’s attack recovers k for *all* 52 nonzero residues $k \in [1, p-1]$ (52/52; an exhaustive implementation validation, Phase 51). The same code path that is inert for $\gcd(n, p) = 1$ is a total break the instant $n = p$.

The dichotomy is exhaustive, not merely sharp. One might ask whether an *intermediate* regime exists—a curve on which $[n]$ partially degenerates on the kernel, $0 \neq \ker([n]|_{\widehat{E}}) \subsetneq \widehat{E}$, giving a graded leak. On $E(\mathbb{F}_p)$ there is none. A partial kernel needs $1 \leq v_p(n) < e-1$ with $v_p(n) \geq 1$, i.e. $p \mid n$; but $n \mid N = \#E(\mathbb{F}_p)$, so $p \mid n$ forces $p \mid N$, and by Hasse $N \in [(\sqrt{p}-1)^2, (\sqrt{p}+1)^2]$ the only multiple of p in range is $N = p$ itself for $p \geq 7$ (we verified no curve with $p \mid N \neq p$ exists for $p = 7, \dots, 23$; the lone exception $N = 2p$ occurs only at $p = 5$). Hence $v_p(n) \in \{0, 1\}$ and $[n]$ acts on $\widehat{E}$ as an automorphism ($p \nmid n$) or, on its p -part, the zero map ($n = p$), never partially. The boundary is a true dichotomy with no curves in between; a “graded inertness theorem” has empty interior here.

9 Numerical verification

Theorem 1 was checked on $y^2 = x^3 + 7$ at $e = 4$ for ten small ordinary curves over $\mathbb{F}_p$, $p = 31, \dots, 163$, all with $\gcd(n, p) = 1$ (Phase 41). These test orders are deliberately small and several are composite— $n \in \{21, 31, 79, 67, 79, 111, 129, 133, 86, 139\}$, five of them composite—chosen for exact-arithmetic reach rather than to model cryptographic-scale orders; since the theorem is regime-agnostic (it needs only $\gcd(n, p) = 1$), they suffice to verify it. (Here n denotes the order of the chosen G and may be a proper divisor of $N = \#E(\mathbb{F}_p)$, e.g. $n = 86$ with $N = 172$ at $p = 157$; the clean set of Table 2 instead requires $n = N$ prime.) Phase 47 additionally verifies every clause on one large-prime-order ordinary curve ($p = 10477$, $n = N = 10639$ prime): C_n again has valuation 1 and additive order exactly p^{e-1} , the antisymmetry is single-valued across all 5319 reflection pairs, and $L = n - 1$ with no degenerate drops. All kernel arithmetic is done through the additive z -coordinate to avoid projective degeneracy on deep kernel points. On every prime: the obstruction $C_n := z([n]\tau(G))$ is nonzero in $p\mathbb{Z}$ of valuation 1 and additive order exactly p^{e-1} (clauses (d),(b)); the antisymmetry $z(\delta(k)) + z(\delta(n - k)) \equiv C_n$ holds for all k (clause (d) / (3)); the integer-representative shift $z(\delta(k + n)) - z(\delta(k)) \equiv C_n$ holds on all eight arithmetically-reliable primes (a group identity; on two primes the deep multiply degenerates in exact arithmetic, flagged by a self-consistency probe); and $\gcd(n, p^{e-1}) = 1$ (clause (c)). The canonical case $\delta_{\text{can}} \equiv 0$ (Lemma 1) was confirmed in an exact p -adic computation. Anomalous recovery is Phase 42 (§8).

10 Discussion

For negative cryptanalysis. The reusable lesson is narrow and practical. (i) A control tests exactly one alternative—the one its null recomputes; before trusting it, enumerate the mechanisms that could generate the statistic and check whether the control separates them. The permutation test here failed because three mechanisms collapsed under it. (ii) Build one null per mechanism, derived from how that mechanism generates the statistic: H1 from the prefix-sum structure (shuffle increments, re-sum), H2 from the reflection’s reach (restrict to a half), and—decisively—a *positive* control synthesizing the surviving mechanism (reflection-only sequences) and showing it reproduces the statistic. (iii) Beware cumulative statistics: any running sum manufactures low-frequency autocorrelation, so “looks autocorrelated” is the null, not the finding. (iv) Where a structural theorem is available it bounds what controls can find; Theorem 1 says no honest control turns δ into an oracle for $\gcd(n, p) = 1$.

For automated and AI-assisted exploration. Pipelines that fan out many probes and surface the most anomalous are false-positive engines: moderate separations are cheap under multiple probing, a real identity against a wrong null produces arbitrarily large ones, and a plausible-sounding control can launder either into a “finding.” The safeguard is not more probes but mechanism-derived nulls, a positive control for any surviving signal, and—where possible—a closed-form reason the probe must be inert. The theorem’s value here is exactly that: it turns “we tried forty things and they failed” into “here is the one-line reason they had to.”

Disclosure. The exploration this paper dissects was itself AI-assisted: the phase scripts, probe variants, and the internal Phase-37 report that certified the anomaly were generated with large-language-model assistance under human direction, and the error was caught at the human audit stage—by interrogating what the permutation control could actually distinguish—not by the pipeline itself. We disclose this because the failure mode documented here is exactly the one such workflows must guard against: an automated analyst that produces both an anomaly and a superfi-

cially appropriate control for it will, with some regularity, produce certified artifacts. The protocol of §11 is written so that it can be imposed on automated searches as a checklist.

Honest scope. Theorem 1 is not new mathematics. It is an explicit instance, in the lift-error language, of the principle of [2, 1, 3] that lifting transports the ECDLP; the anomalous degeneration is Smart’s attack [4]. We claim novelty only for the methodology—the mechanism-derived controls, the positive and converse controls that localize the anomaly, and the closed-form pseudo-variance account of the artifact (Prop. 1)—and for the explicit identification of the obstruction with the antisymmetry constant. The precise boundary of what this refutes is drawn next.

Proof map (prior art / new / empirical). For the reader in a hurry: *prior art*—that lifting transports the ECDLP, that the canonical lift is information-free, and the anomalous (Smart) attack [2, 1, 3, 4, 5, 6]; *new (proved)*—the pseudo-variance classification of U^2 inflation (Prop. 1), its $3/2/0.427\sqrt{L}$ specialization (Prop. 2), and the lift-error packaging of inertness with its sharp $\gcd(n, p) = 1$ boundary (Thm. 1, §8); *new (methodological)*—the mechanism-derived nulls and the positive/converse/sham controls that localize the anomaly to (3); *empirical*—every z -value and ratio in Table 2 and Phases 47–63, plus the carry model (a stated modelling assumption, not proved).

11 What this refutes, what it does not, and how to search next

The attack schema, defined. By an attack in this schema we mean: an adversary, given only the public instance (E, G, Q) , computes a section lift error sequence and a deterministic statistic of it, and uses that statistic to recover k . The crucial point is that the labeled sequence $(k, \delta_\tau(k))$ is an *analyst-side* object—an adversary cannot form it from Q without already knowing k —so the schema is really “offline analysis of a labeled sequence,” and what we ask is whether any section-invariant, publicly computable statistic of it depends on the secret.

The schema this addresses. The result speaks to one specific attack schema: secret extraction from deterministic statistics of section lift errors $\delta_\tau(k)$ —digit, phase, Fourier/Gowers, prefix-sum, spectral, recurrence, or machine-discovered structure in the ordered sequence $(k, \delta_\tau(k))$ —on ordinary curves with $\gcd(n, p) = 1$. By Theorem 1(c,d) and Corollary 1, any such signal computable from (E, G, Q) carries a structural presumption of identity-or-artifact. We state the operative claim in the defensible form: this class of statistics has no *well-defined* secret-dependent content available from public data except at the anomalous boundary; under the standard assumption that the ECDLP is hard on ordinary non-anomalous curves, observed structure in these statistics should be presumed a public identity or an artifact unless it survives section-invariance and mechanism-derived controls. Appendix A is the empirical census of this schema; the theorem is the reason the census came back empty. Table 4 summarizes.

What this does not refute. The claim is bounded. It establishes *nothing* about: nonce-leakage and side-channel attacks on ECDSA; invalid-curve and twist attacks; small-subgroup attacks; small-embedding-degree / pairing (MOV) mechanisms, which are a separate functional inert here only for the independent reason of huge embedding degree; quantum algorithms; deliberately weakened curves; extension-field cases $E/\mathbb{F}_{p^m}$, where the Hasse dichotomy of §8 can fail; and—most relevant for the lifting literature—non-section-based global lift constructions. We claim no impossibility theorem for all p -adic or global lifts; in Silverman’s “four faces” taxonomy [2, 1] the killed schema

Killed schema	Why inert	Boundary / not covered
Statistics of $\delta_\tau(k)$ and its digit/phase/Fourier encodings, ordinary $\gcd(n, p) = 1$	Thm. 1: secret part $[k]c$ not publicly determined; visible δ is identity-or-artifact ((3))	$n = p$: Smart’s attack (§8). Not covered: side channels, pairings, twists, quantum, $\mathbb{F}_{p^m}$, non-section global lifts

Table 4: Scope of the refutation. The schema is structurally dead on ordinary curves; the reusable contribution is the mechanism-resolved falsification protocol, not a broad impossibility claim.

is one cell, not the map, and the anomalous boundary is precisely where a genuine p -adic attack (Smart) lives—we recover it, we do not contradict it.

The Mechanism-Resolved Null Protocol. A credible new direction in this area should, before experimentation: (i) name the public invariant it computes from (E, G, Q) ; (ii) show it is well-defined independent of the section choice (the Theorem 1(d) test); (iii) show it is not forced to be canonically zero (Lemma 1) or a public identity (the fate of (3)); (iv) pair every candidate signal with mechanism-derived nulls—one for summation/walk artifacts, one for algebraic symmetries (reflection foremost), one for automorphism and coordinate-choice effects—*plus* a positive control synthesizing the surviving mechanism and, where possible, the converse test of breaking it (and the sham test of perturbing without breaking it, §5); and (v) treat as evidence only a signal that survives all of these, scales on large ordinary prime-order curves, and admits a structural explanation not reducible to known lift inertness. The growth law (5) is a ready discriminator: $\sqrt{n}$ growth toward the $3/2$ ceiling is the fingerprint of an index-reversal identity, not of an accumulating leak. It also serves as an experiment-design tool: given a target residual size, the $0.427\sqrt{L}$ law fixes in advance the L at which the equivalence bound of §6(ii) would exclude it.

Where a genuine signal could still live. Three directions are suggested, not foreclosed, by this work. First, the pseudo-variance classification (Prop. 1) is not specific to the ECDLP: estimating the per-mode pseudo-variance spectrum of any anomalous fourth-moment statistic identifies the linear index constraint responsible, and extending the dictionary to higher Gowers norms and multidimensional Fourier would make it a general-purpose audit for automated search pipelines. Second, the exhaustive dichotomy of §8 is a prime-field fact: over $\mathbb{F}_{p^m}$ and on higher-dimensional abelian varieties the kernel of reduction has higher rank and $[n]$ *can* degenerate partially, so the “graded leak” regime that is provably empty here has nonempty interior there—a principled place to search, with the protocol above imposed from the start. Third, the two identities that survived this exploration (automorphism-equivariance and the antisymmetry) invite a completeness question: characterize *all* public linear constraints on δ and show the U^2 signature is determined by that list, upgrading “every probe failed” to “here is everything any probe can see.”

12 Reproducibility

Because the paper’s thesis is that controls can mislead, its own workflow is held to a higher standard. All numbers are regenerated by the repository at <https://github.com/wtrevena/ecdsa-lift>, commit 566a48939199cb870bc9c069696063008130dd79, release tag `paper-v4` (an archival DOI will accompany the camera-ready); Python 3.10, NumPy 2.2, SymPy 1.14, Matplotlib 3.10. Each phase is a standalone script under `experiments/` writing a JSON record under `results/`. Minimal

reproduction of the headline claims: `python3 experiments/phase43_revision_controls.py` re-generates Table 2 (about two minutes on commodity hardware), `python3 experiments/phase61_mass_variance.py` the Proposition 2 constants (under one minute), and `python3 experiments/phase62_figure.py` Figure 1 (seconds); `run_all.sh` (exact filename, underscore) runs every phase from the fixed seed 20260609 in under an hour on a single core. The pre-registered protocol (§6) was fixed before the clean re-run. Results fixed at first submission are Phases 1–43 and 47–48; the converse, non-CM, all- k , calibration, and closed-form phases added in the first revision are 49–59; the genuine-section converse, the mass-variance derivation, and the figure added in this revision are 60–62. All phases, including the exploratory 44–46 (which no argument in this paper uses), are frozen at the commit above; nothing is reported from outside it.

13 Conclusion

A long negative exploration produced one seductive anomaly. A standard control endorsed it; mechanism-derived controls, capped by a positive control, killed it and attributed it entirely to an exact public identity; and an inertness theorem explains why no such anomaly on ordinary curves can be more than artifact-or-identity, with the boundary falling exactly at the anomalous curves where the lift attack genuinely works. The cryptanalytic verdict is the expected negative one. The transferable result is the Mechanism-Resolved Null Protocol: name your alternative hypotheses, build a null for each from its mechanism, synthesize the survivor to confirm it, break it to watch the signal die (and perturb without breaking it to watch the signal survive), and prefer a structural reason for inertness over an accumulation of passed tests.

A Phase index

Phase(s)	Target	Verdict
1–2	injectivity / pseudorandomness of δ	injective, pseudorandom
3–4	Smart–SSSA; homomorphic section from H^2	section exists, attack does not
5–6	CM-linearization; HNP on cocycle	pseudorandom; no bias
7–10	MOV; poly fit; LLL coords; ord/ss bridge	inapplicable / no leak
11–13	genus-2 Jacobian; GLV-CRT; Weil pairing	no new lever
14	$\text{Aut}(E)$ -equivariance rank law of δ -digit	public, no advantage
15–20	triple cocycle; DFT; Berlekamp–Massey; valuation; mult.	all inert
21b	antisymmetry $\delta(k) + \delta(n - k) = [n]\tau(G)$	exact, public
22–29	coord change; formal log; Mazur–Tate; Mahler	no structure
30–36	Sage validation; 2-cocycle; canonical lift; Iwasawa	inert
37	Gowers U^2 anomaly	false positive (this paper)
40,40b	increment-shuffle and half-sequence controls	localizes to 21b
41	numerical verification of Theorem 1	confirmed
42	anomalous boundary (Smart)	52/52 recovery
43	clean-ordinary re-run + positive control	confirms §5
44–46	aggregation/MI estimators; alt. uniformizers; OEIS scan	exploratory; frozen, unused here
47	large-prime-order confirmatory curve ($p = 10477$)	confirms §5, §9
48	U^2 growth law (Prop. 2: $3/2$ mass, $\sqrt{n}$)	artifact signature
49,49b	first converse attempt (non-section rule; confound documented)	superseded by 60
50	non-CM / $j=1728$ curves	pattern CM-independent
51	Smart attack, all $k \in [1, p-1]$	52/52 recovery
52–53	TOST equivalence; half-sequence power	residual/power bounded
54	look-elsewhere battery on surrogates	5σ is cheap
55–56	fixed-shift discriminator; large- L (10^6) constant	1.001 vs 1.499; 0.427
57	pseudo-variance classification (Prop. 1)	$\mathbb{E} Z ^4 = 2\sigma^4 + \rho ^2$
58	carry model q_j ; resolvability $\sim \sqrt{L}$	prediction at $L \gtrsim 3000$
59	U^3 inflation $\rightarrow 1$ (excess $\sim 1/L$)	confirms baseline
60	genuine kernel-offset asymmetric section (converse + sham)	collapse iff oddness broken
61	U^2 mass variance: Dirichlet $4/L^3$, delta method	closes Prop. 2
62	figure generation from committed results	Fig. 1
63	conjugation & finite-alphabet checks (Props. 1–2)	per-mode $ \rho /\sigma^2=1$; alphabet-free

Table 5: Index of the exploration. Full scripts, fixed seeds, and JSON outputs are in the `ecdsa-lift` repository.

B CM-independence: $j=1728$ and non-CM curves

The clean set of Table 2 is $j=0$ ($y^2 = x^3 + 7$), which has extra automorphisms. Table 6 repeats the protocol on a $j=1728$ curve and three generic non-CM curves; the $3/2$ inflation and the real-vs-synthetic equivalence reproduce regardless of CM (Phase 50), confirming the mechanism is the antisymmetry and not the CM structure.

C Null-model pseudocode

Each control recomputes U^2 under a different null; the input is the real sequence $s(k) = z(\delta(k))$, $k = 1, \dots, n-1$, and digit j .

p	j	n	L	real/inc	syn/rand	real/syn
181	1728	101	100	+2.6..4.9	+4.1..4.4	-0.8..0.3
163	6	173	172	+4.2..5.1	+5.3..5.5	-0.3..-0.1
521	316	479	478	+6.6..11.8	+9.6..9.8	-1.4..1.3
653	446	613	612	+6.7..11.3	+10.2..10.3	-2.1..0.5

Table 6: Phase 50, z -ranges over digits 1, 2, 3. Each curve has prime generator order n with $\gcd(n, p) = 1$; the $j=1728$ row uses the largest prime-order subgroup ($y^2=x^3+ax$ always has even $\#E$). The reflection signature and the positive-control match are CM-independent.

- **Value permutation:** apply a uniform random permutation π to the multiset $\{s(k)\}$, re-index, recompute U^2 . (Tests “value distribution vs. ordering”; cannot separate H1/H2/H3.)
- **Increment shuffle (H1):** $d(k) = s(k+1) - s(k)$; draw a random permutation of $\{d(k)\}$, re-sum to a new walk s' , recompute U^2 . (Same increment multiset, only order changes.)
- **Half-sequence (H2):** restrict to $k = 1, \dots, \lfloor (n-1)/2 \rfloor$ (no reflection partner in-window), recompute the increment-shuffle comparison at half length.
- **Reflection-only synthetic (positive):** draw the free half $s(k)$ i.i.d. uniform, set $s(n-k) = C - s(k)$ with $C = z([n]\tau(G))$, encode digit j , recompute U^2 .
- **Kernel-offset converse / sham:** form $\tau'(P) = \tau(P) + [h(P)]K_1$ with $K_1 = [n]\tau(G)$; set $z(\delta'(k)) = z(\delta(k)) + (k h(G) - h(kG))C_n \bmod p^e$; encode and recompute. Converse: h non-odd and full-entropy $(x, x+7y)$. Sham: h odd (y) .

The empirical null for each is 300 draws; “ z ” is the standardized gap of the real statistic against that null’s mean and SD.

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